LEFT DUALS IN THE DERIVED CATEGORY ARE PERFECT AND TERMWISE DUAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- K-flat resolutions. Cohomology of Sheaves, Lemma cohomology-lemma-K-flat-resolution. K-flat resolution of the left dual with flat terms.
- Local representability of derived maps. Cohomology of Sheaves, Lemma cohomology-lemma-local-actual. sigma as an actual map of complexes, null-homotopy.
- Precise perfection criterion. Cohomology of Sheaves, Lemma cohomology-lemma-perfect-precise. Concludes perfection from the descent.
- Maps from complexes concentrated in negative degrees. Derived Categories, Lemma derived-lemma-negative-exts. L -> E^*[1] as a map of complexes.

Source reference. Original reference 0FFP: sources/stacks-chow/tags.txt, source lines 19294–19294. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0FPC: sources/stacks-chow/tags.txt, source lines 19597–19597. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 06YF: sources/stacks-chow/tags.txt, source lines 8884–8884. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 05NI: sources/stacks-chow/tags.txt, source lines 7307–7307. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 05T7: sources/stacks-chow/tags.txt, source lines 7471–7471. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 08CP: sources/stacks-chow/tags.txt, source lines 10609–10609. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 06XS: sources/stacks-chow/tags.txt, source lines 8861–8861. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Date: stacks-project:cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 08C9: sources/stacks-chow/tags.txt, source lines 10594–10594. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. Let M be an object of $D(\mathcal{O}_X)$. If M has a left dual in the monoidal category $D(\mathcal{O}_X)$ (Categories, Definition 0FFP) then M is perfect and the left dual is as constructed in Example 0FPC.*

Proof. Let $x \in X$. It suffices to find an open neighbourhood U of x such that M restricts to a perfect complex over U. Hence during the proof we can (finitely often) replace X by an open neighbourhood of x. Let N, η, ϵ be a left dual.

We are going to use the following argument several times. Choose any complex $\mathcal{M}^\bullet$ of $\mathcal{O}_X$-modules representing M. Choose a K-flat complex $\mathcal{N}^\bullet$ representing N whose terms are flat $\mathcal{O}_X$-modules, see Lemma 06YF. Consider the map

$$\eta : \mathcal{O}_X \to \mathrm{Tot}(\mathcal{M}^\bullet \otimes_{\mathcal{O}_X} \mathcal{N}^\bullet)$$

After shrinking X we can find an integer N and for $i = 1, \ldots, N$ integers $n_i \in \mathbf{Z}$ and sections f_i and g_i of $\mathcal{M}^{n_i}$ and $\mathcal{N}^{-n_i}$ such that

$$\eta(1) = \sum\nolimits_i f_i \otimes g_i$$

Let $\mathcal{K}^\bullet \subset \mathcal{M}^\bullet$ be any subcomplex of $\mathcal{O}_X$-modules containing the sections f_i for $i = 1, \ldots, N$. Since $\mathrm{Tot}(\mathcal{K}^\bullet \otimes_{\mathcal{O}_X} \mathcal{N}^\bullet) \subset \mathrm{Tot}(\mathcal{M}^\bullet \otimes_{\mathcal{O}_X} \mathcal{N}^\bullet)$ by flatness of the modules $\mathcal{N}^n$, we see that η factors through

$$\tilde{\eta} : \mathcal{O}_X \to \mathrm{Tot}(\mathcal{K}^\bullet \otimes_{\mathcal{O}_X} \mathcal{N}^\bullet)$$

Denoting K the object of $D(\mathcal{O}_X)$ represented by $\mathcal{K}^\bullet$ we find a commutative diagram

$$\begin{array}{ccccc} M & \xrightarrow{\eta\otimes 1} & M \otimes^{\mathbf{L}} N \otimes^{\mathbf{L}} M & \xrightarrow{1\otimes\epsilon} & M \\ & \searrow^{\tilde{\eta}\otimes 1} & \uparrow & & \uparrow \\ & & K \otimes^{\mathbf{L}} N \otimes^{\mathbf{L}} M & \xrightarrow{1\otimes\epsilon} & K \end{array}$$

Since the composition of the upper row is the identity on M we conclude that M is a direct summand of K in $D(\mathcal{O}_X)$.

As a first use of the argument above, we can choose the subcomplex $\mathcal{K}^\bullet = \sigma_{\geq a}\tau_{\leq b}\mathcal{M}^\bullet$ with $a < n_i < b$ for $i = 1, \ldots, N$. Thus M is a direct summand in $D(\mathcal{O}_X)$ of a bounded complex and we conclude we may assume M is in $D^b(\mathcal{O}_X)$. (Recall that the process above involves shrinking X.)

Since M is in $D^b(\mathcal{O}_X)$ we may choose $\mathcal{M}^\bullet$ to be a bounded above complex of flat modules (by Modules, Lemma 05NI and Derived Categories, Lemma 05T7). Then we can choose $\mathcal{K}^\bullet = \sigma_{\geq a}\mathcal{M}^\bullet$ with $a < n_i$ for $i = 1, \ldots, N$ in the argument above. Thus we find that we may assume M is a direct summand in $D(\mathcal{O}_X)$ of a bounded complex of flat modules. In particular, M has finite tor amplitude.

Say M has tor amplitude in $[a, b]$. Assuming M is m-pseudo-coherent we are going to show that (after shrinking X) we may assume M is $(m-1)$-pseudo-coherent. This will finish the proof by Lemma 08CP and the fact that M is $(b+1)$-pseudo-coherent in any case. After shrinking X we may assume there exists a strictly perfect complex $\mathcal{E}^\bullet$ and a map $\alpha : \mathcal{E}^\bullet \to M$ in $D(\mathcal{O}_X)$ such that $H^i(\alpha)$ is an isomorphism for $i > m$ and surjective for $i = m$. We may and do assume that $\mathcal{E}^i = 0$ for $i < m$. Choose a distinguished triangle

$$\mathcal{E}^\bullet \to M \to L \to \mathcal{E}^\bullet[1]$$

Observe that $H^i(L) = 0$ for $i \geq m$. Thus we may represent L by a complex $\mathcal{L}^\bullet$ with $\mathcal{L}^i = 0$ for $i \geq m$. The map $L \to \mathcal{E}^\bullet[1]$ is given by a map of complexes $\mathcal{L}^\bullet \to \mathcal{E}^\bullet[1]$ which is zero in all degrees except in degree $m-1$ where we obtain a map $\mathcal{L}^{m-1} \to \mathcal{E}^m$, see Derived Categories, Lemma 06XS. Then M is represented by the complex

$$\mathcal{M}^\bullet : \ldots \to \mathcal{L}^{m-2} \to \mathcal{L}^{m-1} \to \mathcal{E}^m \to \mathcal{E}^{m+1} \to \ldots$$

Apply the discussion in the second paragraph to this complex to get sections f_i of $\mathcal{M}^{n_i}$ for $i = 1, \ldots, N$. For $n < m$ let $\mathcal{K}^n \subset \mathcal{L}^n$ be the $\mathcal{O}_X$-submodule generated by the sections f_i for $n_i = n$ and $d(f_i)$ for $n_i = n-1$. For $n \geq m$ set $\mathcal{K}^n = \mathcal{E}^n$. Clearly, we have a morphism of distinguished triangles

$$\begin{array}{ccccccc}
\mathcal{E}^\bullet & \longrightarrow & \mathcal{M}^\bullet & \longrightarrow & \mathcal{L}^\bullet & \longrightarrow & \mathcal{E}^\bullet[1] \\
\uparrow & & \uparrow & & \uparrow & & \uparrow \\
\mathcal{E}^\bullet & \longrightarrow & \mathcal{K}^\bullet & \longrightarrow & \sigma_{\leq m-1}\mathcal{K}^\bullet & \longrightarrow & \mathcal{E}^\bullet[1]
\end{array}$$

where all the morphisms are as indicated above. Denote K the object of $D(\mathcal{O}_X)$ corresponding to the complex $\mathcal{K}^\bullet$. By the arguments in the second paragraph of the proof we obtain a morphism $s : M \to K$ in $D(\mathcal{O}_X)$ such that the composition $M \to K \to M$ is the identity on M. We don't know that the diagram

$$\begin{array}{ccccc}
\mathcal{E}^\bullet & \longrightarrow & \mathcal{K}^\bullet & = & K \\
{\scriptstyle \mathrm{id}}\uparrow & & & & \uparrow{\scriptstyle s} \\
\mathcal{E}^\bullet & \xrightarrow{i} & \mathcal{M}^\bullet & = & M
\end{array}$$

commutes, but we do know it commutes after composing with the map $K \to M$. By Lemma 08C9 after shrinking X we may assume that $s \circ i$ is given by a map of complexes $\sigma : \mathcal{E}^\bullet \to \mathcal{K}^\bullet$. By the same lemma we may assume the composition of σ with the inclusion $\mathcal{K}^\bullet \subset \mathcal{M}^\bullet$ is homotopic to zero by some homotopy $\{h^i : \mathcal{E}^i \to \mathcal{M}^{i-1}\}$. Thus, after replacing $\mathcal{K}^{m-1}$ by $\mathcal{K}^{m-1} + \mathrm{Im}(h^m)$ (note that after doing this it is still the case that $\mathcal{K}^{m-1}$ is generated by finitely many global sections), we see that σ itself is homotopic to zero! This means that we have a commutative solid diagram

$$\begin{array}{ccccccc}
\mathcal{E}^\bullet & \longrightarrow & M & \longrightarrow & \mathcal{L}^\bullet & \longrightarrow & \mathcal{E}^\bullet[1] \\
\uparrow & & \uparrow & & \uparrow & & \uparrow \\
\mathcal{E}^\bullet & \longrightarrow & K & \longrightarrow & \sigma_{\leq m-1}\mathcal{K}^\bullet & \longrightarrow & \mathcal{E}^\bullet[1] \\
\uparrow & & \uparrow{\scriptstyle s} & & \uparrow & & \uparrow \\
\mathcal{E}^\bullet & \longrightarrow & M & \longrightarrow & \mathcal{L}^\bullet & \longrightarrow & \mathcal{E}^\bullet[1]
\end{array}$$

By the axioms of triangulated categories we obtain a dotted arrow fitting into the diagram. Looking at cohomology sheaves in degree $m-1$ we see that we obtain

$$\begin{array}{ccccc}
H^{m-1}(M) & \longrightarrow & H^{m-1}(\mathcal{L}^\bullet) & \longrightarrow & H^m(\mathcal{E}^\bullet) \\
\uparrow & & \uparrow & & \uparrow \\
H^{m-1}(K) & \longrightarrow & H^{m-1}(\sigma_{\leq m-1}\mathcal{K}^\bullet) & \longrightarrow & H^m(\mathcal{E}^\bullet) \\
\uparrow & & \uparrow & & \uparrow \\
H^{m-1}(M) & \longrightarrow & H^{m-1}(\mathcal{L}^\bullet) & \longrightarrow & H^m(\mathcal{E}^\bullet)
\end{array}$$

Since the vertical compositions are the identity in both the left and right column, we conclude the vertical composition $H^{m-1}(\mathcal{L}^\bullet) \to H^{m-1}(\sigma_{\leq m-1}\mathcal{K}^\bullet) \to H^{m-1}(\mathcal{L}^\bullet)$ in the middle is surjective! In particular $H^{m-1}(\sigma_{\leq m-1}\mathcal{K}^\bullet) \to H^{m-1}(\mathcal{L}^\bullet)$ is surjective. Using the induced map of long exact sequences of cohomology sheaves from the morphism of triangles above, a diagram chase shows this implies $H^i(K) \to H^i(M)$ is an isomorphism for $i \geq m$ and surjective for $i = m - 1$. By construction we can choose an $r \geq 0$ and a surjection $\mathcal{O}_X^{\oplus r} \to \mathcal{K}^{m-1}$. Then the composition

$$(\mathcal{O}_X^{\oplus r} \to \mathcal{E}^m \to \mathcal{E}^{m+1} \to \ldots) \longrightarrow K \longrightarrow M$$

induces an isomorphism on cohomology sheaves in degrees $\geq m$ and a surjection in degree $m - 1$ and the proof is complete. $\square$